

Heterogeneous Graph Neural Networks for Transaction-Level Fraud Detection

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Abstract

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(~300 words)
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Chapter 1

Introduction

1.1 Motivation

Financial fraud imposes enormous costs on banks, customers, and society. Traditional rule-based transaction monitoring systems generate false positive rates between 95 and 98 percent (Gerlings and Constantiou, 2023; Cardoso et al., 2022), meaning that investigators spend the vast majority of their time reviewing alerts that turn out to be legitimate. This inefficiency creates both operational cost and detection risk: while investigators are occupied with false positives, genuine fraud may go undetected.

The structure of financial transaction data suggests that graph-based approaches may offer a path forward. Transactions naturally form a network connecting senders, receivers, and intermediaries, and fraudulent activity often manifests as anomalous patterns in this network rather than in individual transaction attributes. Prior work has confirmed that graph signal is present in fraud data: accounts in the immediate neighbourhood of known fraud are significantly more likely to be fraud senders themselves, and a substantial fraction of fraud senders are repeat offenders (Johannessen and Jullum, 2023).

Real banking environments are inherently heterogeneous, containing multiple entity types connected through diverse relationship types. Homogeneous graph representations that treat all nodes and edges as identical discard this structural information. Heterogeneous graph neural networks (GNNs) preserve type-specific distinctions, and recent work has shown their value for fraud detection at scale (Hu et al., 2020; Rao et al., 2022; Johannessen and Jullum, 2023). However, these methods rely entirely on labeled data—a resource that is scarce and inherently incomplete in the fraud domain, since labels reflect only *detected* fraud.

Self-supervised graph learning offers a complementary strategy. By learning representations from the full unlabeled transaction graph, self-supervised methods can capture structural patterns that supervised approaches miss due to label scarcity. Cardoso et al. (2022) demonstrated this with LaundroGraph on a homogeneous graph, but the combina-

tion of self-supervised pretraining with heterogeneous graph structure remains unexplored for fraud detection.

1.2 Research Questions and Hypotheses

This thesis investigates the following hypotheses:

H1 — Heterogeneous graph structure improves detection quality.

Heterogeneous GNN architectures achieve a more favourable precision-recall tradeoff than conventional supervised (tabular) models, surfacing more true fraud cases per investigator-hour spent reviewing alerts.

H2 — Self-supervised pretraining yields useful representations despite incomplete labels.

A self-supervised approach (HGMAE) learning the structural patterns of “normal” transaction behaviour produces embeddings that generalise to the detection of anomalous fraud patterns—even accounting for the fact that ground truth labels are inherently incomplete.

H3 — Self-supervised heterogeneous embeddings improve the precision-recall tradeoff under label scarcity.

Combining self-supervised pretraining (HGMAE) with a downstream heterogeneous classifier (HMPNN or HGT) yields a better operational tradeoff than either approach alone—particularly in settings where labeled fraud cases are scarce and ground truth is incomplete.

H4 — GNN explanations provide more actionable investigator guidance than tabular feature importance.

Graph-native explainability methods (e.g., attention weights from HGT) surface the specific transaction paths and counterparty relationships that constitute suspicious behaviour—information that per-feature importance scores from tree-based models cannot structurally represent.

1.3 Contributions

This thesis makes the following contributions:

1. A systematic experimental ladder (L0–L4) that isolates the marginal contribution of graph structure, heterogeneous typing, and self-supervised pretraining to fraud detection performance on real banking data.

2. The first application of Heterogeneous Graph Masked Autoencoders (HGMAE) to financial transaction data, bridging the gap between self-supervised graph learning (previously limited to homogeneous graphs in this domain) and heterogeneous graph modelling.
3. A comparison of three heterogeneous graph topologies (V1, V2, TXN_V1) encoding different levels of structural detail from the same underlying transaction data.
4. An evaluation framework grounded in operational metrics (PR-AUC, false positives per true positive) rather than accuracy, reflecting the real-world constraints of fraud detection.

1.4 Thesis Outline

The remainder of this thesis is structured as follows:

Chapter 2

reviews the relevant literature on graph-based fraud detection, heterogeneous GNNs, self-supervised graph learning, and evaluation under class imbalance.

Chapter 3

describes the dataset, feature engineering, graph construction, and temporal splitting strategy.

Chapter 4

presents the experimental ladder and the models evaluated at each level.

Chapter 5

reports the experimental results.

Chapter 6

discusses the findings in relation to the hypotheses and prior work.

Chapter 7

concludes with a summary and directions for future work.

Chapter 2

Background and Related Work

This chapter situates the thesis within the broader landscape of graph-based fraud detection research. We begin with the fraud detection problem in banking, trace the evolution from tabular machine learning through graph representations to heterogeneous graph neural networks, and conclude by identifying the research gap that this thesis addresses.

2.1 Transaction Fraud Detection

Fraud in retail banking remains one of the most persistent challenges facing financial institutions. Vallarino (2025) estimates that financial fraud causes global losses exceeding 1.8 trillion dollars annually, with traditional systems unable to keep pace with increasingly sophisticated schemes. Banks are legally obligated to monitor transactions and flag suspicious activity, but the rule-based systems that underpin most transaction monitoring frameworks are notoriously imprecise. Gerlings and Constantiou (2023), in a two-year action research study conducted with a European bank, found that traditional rule-based scenarios generate false positive rates between 95 and 98 percent, meaning that the vast majority of alerts reviewed by investigators are ultimately unproductive. Cardoso et al. (2022) corroborate this figure, noting that legacy systems routinely exceed 95 percent false positive rates.

The operational cost of these false positives is substantial: investigators must manually review each flagged alert within tight regulatory timeframes, and the sheer volume of unproductive alerts diverts resources away from genuinely suspicious activity. This inefficiency has motivated a sustained effort to apply machine learning methods to fraud detection, first through tabular approaches and more recently through graph-based architectures that can capture the relational structure inherent in financial transaction data.

2.2 Traditional Machine Learning for Fraud Detection

Early applications of machine learning to fraud detection relied on supervised tabular models. Random forests, gradient-boosted trees such as XGBoost, and logistic regression have been widely deployed for transaction-level classification, typically operating on hand-engineered features derived from transaction amounts, frequencies, counterparty statistics, and temporal patterns (Cheng et al., 2024). Bathala et al. (2024) demonstrated the use of deep feedforward neural networks with SHAP-based explainability on synthetic transaction data, achieving weighted F1 scores around 0.91 but acknowledging that synthetic datasets may not capture real-world complexity.

A fundamental limitation of all tabular approaches, however, is that they treat each transaction or customer as an independent observation, discarding the rich relational context in which financial activity occurs. A single suspicious transaction may appear benign in isolation but reveal its fraudulent nature only when examined alongside the network of accounts, counterparties, and intermediary entities through which funds have flowed. This structural blindness—the inability to reason about the topology of financial relationships—has driven the field toward graph-based representations.

2.3 Graph Representations of Financial Data

The recognition that financial fraud is fundamentally a relational phenomenon has motivated a shift toward graph-based representations of transaction data. As Johannessen and Jullum (2023) argue, fraudulent activity often involves groups of actors collaborating to move funds—and the patterns that betray this collaboration are often invisible when customers or transactions are examined in isolation. Network analysis can uncover circumstances that would be impossible to detect through a purely entity-based approach, simply because the necessary relational information is not present in tabular feature sets. At the simplest level, accounts can be modeled as nodes and transactions as directed edges, forming a homogeneous transaction graph. However, this representation discards critical distinctions between different types of entities and relationships.

Real banking environments are inherently heterogeneous. A bank’s data contains multiple entity types—individual customers, corporate customers, external accounts, merchants, devices, geographic locations—connected through diverse relationship types such as financial transfers, shared ownership, business ties, and common attributes. Singh et al. (2024) emphasize that financial data encompasses diverse entities like cardholders, merchants, and transactions, each with distinct attributes and relationships that homogeneous graphs cannot fully capture. A heterogeneous graph, by contrast, preserves these distinctions: nodes carry type-specific feature sets, and edges encode both the nature and the properties of each relationship. Johannessen and Jullum (2023) further note that

existing heterogeneous GNN methods such as RGCN (Schlichtkrull et al., 2018), HAN (Wang et al., 2019), and HGT (Hu et al., 2020) were not designed to incorporate edge features—yet transaction properties like amounts, timestamps, and currency types carry essential information for effective fraud detection.

Yu and colleagues (2024) explored knowledge graph construction for fraud detection, representing bank accounts as nodes and transactions as edges with attributes including amount, currency, and payment format. They applied Graph Attention Networks with focal loss to address class imbalance on a synthetic IBM dataset of 1,243 transactions, finding that timestamp and payment format were the most discriminative features. However, performance degraded as test set size increased—accuracy dropped from 0.70 to 0.65 when scaling from 2,000 to 4,000 records—an important cautionary note about generalization on small-scale synthetic data.

Once a graph representation is constructed, there are broadly two strategies for incorporating its information into a detection model. The first is to *explicitly* compute graph-derived features—such as centrality metrics, random walk statistics, or neighborhood aggregates—and feed them as additional inputs to a conventional classifier. Naser Eddin et al. (2021) exemplify this approach, extracting degree statistics and GuiltyWalker propagation scores from a dynamic transaction graph and using them alongside entity-level features in a LightGBM model. While effective, such hand-crafted features may not capture the structural patterns most informative for the downstream task, and their design requires considerable domain expertise. The second strategy is to *implicitly* learn representations through graph neural networks that operate directly on the graph structure, jointly optimizing feature extraction and classification in an end-to-end fashion. As Cardoso et al. (2022) note, implicitly derived representations through node embedding approaches have generally achieved better downstream classification results than curated graph features, supporting the argument that end-to-end learned representations contain additional, meaningful information beyond what manual feature engineering can capture.

2.4 Graph Neural Networks

The application of graph neural networks to financial fraud detection has grown rapidly in recent years. Cheng et al. (2024) provide a comprehensive review of over 100 studies, organizing GNN approaches for fraud detection into four categories: graph convolutional networks (GCNs), graph attention networks (GATs), graph temporal networks, and heterogeneous graph neural networks. GCNs, introduced by Kipf and Welling (2017), aggregate neighborhood information through spectral convolutions on the graph Laplacian, while GraphSAGE (Hamilton et al., 2017) extends this to an inductive setting through neighborhood sampling and aggregation. These homogeneous architectures laid the groundwork for graph-based fraud detection but treat all nodes and edges as identical,

which is a poor fit for the multi-relational structure of financial networks. Cheng et al. (2024) further note that GCNs suffer from over-smoothing as depth increases, losing the distinct node characteristics that are essential for distinguishing fraudulent from legitimate behavior.

2.5 Knowledge Graph Embeddings

An alternative to message-passing GNNs is to learn embeddings through knowledge graph embedding (KGE) methods, which model relational data as triples (h, r, t) representing a head entity, a relation type, and a tail entity.

TransE (Bordes et al., 2013) learns embeddings such that $\mathbf{h} + \mathbf{r} \approx \mathbf{t}$ when the triple (h, r, t) holds, using a margin-based ranking loss. DistMult (Yang et al., 2015) instead scores triples via a bilinear function $\mathbf{h}^\top \text{diag}(\mathbf{r}) \mathbf{t}$, allowing richer interaction between entity embeddings. Both methods produce entity and relation embeddings that can be used for downstream classification.

In the context of fraud detection, KGE methods offer a way to learn representations of accounts and transaction types without explicit message passing. However, they do not naturally incorporate rich edge features (transaction amounts, timestamps) and are limited to capturing structural patterns encoded in the graph topology. This thesis includes TransE and DistMult as Level 2 baselines to assess whether structural embeddings alone carry useful fraud signal.

2.6 Heterogeneous Graph Neural Networks

Recognizing the limitations of homogeneous approaches, a line of work has developed GNN architectures that explicitly model heterogeneous node and edge types.

2.6.1 Heterogeneous Graph Transformer (HGT)

The Heterogeneous Graph Transformer, proposed by Hu et al. (2020), introduced node-type and edge-type dependent attention mechanisms for heterogeneous graphs. HGT uses separate key, query, and value projections parameterized by meta-relation triplets—source node type, edge type, and target node type—allowing the model to learn type-specific importance weights during message passing. On the Open Academic Graph (179 million nodes, 2 billion edges), HGT outperformed prior methods by 9 to 21 percent across downstream tasks, establishing the viability of heterogeneous attention at web scale.

Rao et al. (2022) built directly on HGT in developing xFraud, an explainable fraud detection system deployed at eBay. The xFraud detector constructs a heterogeneous graph with transaction nodes linked to buyers, sellers, payment tokens, email addresses, and

shipping addresses, then applies heterogeneous mutual attention and message passing with self-attention mechanisms. On eBay’s industrial-scale dataset (1.1 billion nodes, 3.7 billion edges), xFraud achieved an AUC of 0.91, outperforming both vanilla GCN-based approaches (AUC 0.90) and standard GAT (AUC 0.89). Notably, xFraud replaced HGT’s original HGSampling with a GraphSAGE-style sampler for efficiency on sparse financial graphs, reducing inference time while maintaining detection quality.

2.6.2 Heterogeneous Message Passing Neural Network (HMPNN)

Johannessen and Jullum (2023) presented the first published application of GNNs to large-scale real-world heterogeneous bank data from DNB, Norway’s largest financial institution. Their Heterogeneous Message Passing Neural Network extends standard MPNNs by introducing edge-type-specific message functions that incorporate edge features—a critical design choice, since transaction properties such as amounts, timestamps, and currency types carry essential information that purely structural approaches would miss. On a graph of over 5 million nodes and 10 million edges spanning individual customers, organization customers, and external accounts connected through financial transactions and business ties, HMPNN demonstrated that heterogeneous graph structure preserves information that homogeneous approaches lose.

2.6.3 Other Heterogeneous Approaches

Singh et al. (2024) proposed a heterogeneous graph autoencoder for credit card fraud detection, constructing a tripartite graph of cardholders, merchants, and transactions. Their autoencoder, trained exclusively on legitimate transactions, flags anomalous reconstruction errors as potential fraud, achieving an AUC-PR of 0.89 and an F1 score of 0.81. The autoencoder paradigm is notable because it sidesteps the label scarcity problem by learning what constitutes normal behavior rather than requiring explicit fraud labels.

Stan (2024), in an MSc thesis at Utrecht University, applied provably powerful directed multigraph GNNs on a large-scale synthetic transaction dataset of approximately 9.5 million transactions with a 0.14 percent illicit rate, introducing ego IDs, heterogeneous message passing, and port numbering to enhance expressiveness beyond what standard message-passing GNNs can capture.

2.7 Self-Supervised Learning on Graphs

While the studies reviewed above demonstrate the value of heterogeneous graph structure for fraud detection, they share a common dependence on labeled data—a resource that is particularly scarce and unreliable in the fraud detection domain. Ground truth labels are inherently incomplete: they reflect only detected and confirmed fraud, not the true

underlying illicit activity. This means that supervised models are trained on a biased and incomplete view of the positive class, and the negative class (ostensibly legitimate transactions) is contaminated by undetected fraud. A parallel line of research has therefore explored self-supervised graph learning as a strategy for overcoming this severe label scarcity.

2.7.1 LaundroGraph

[Cardoso et al. \(2022\)](#) introduced LaundroGraph, the first fully self-supervised system for fraud detection on transaction graphs. Operating on a customer-transaction bipartite graph from a real European bank (3.2 million customer nodes and 17.7 million transaction nodes), LaundroGraph uses a three-layer GAT encoder with self-supervised link prediction to learn node embeddings, then derives anomaly scores from reconstruction errors on held-out transactions. LaundroGraph outperformed non-graph baselines by 12 percentage points in AUC (94.83 percent versus 82.58 percent for LightGBM), and also surpassed the contrastive self-supervised baseline Deep Graph Infomax by nearly 9 percentage points. However, LaundroGraph operates on a homogeneous bipartite graph with only two node types and a single edge type, leaving open the question of whether richer heterogeneous graph structures could further improve self-supervised representations.

2.7.2 Heterogeneous Graph Masked Autoencoders (HGMAE)

The Heterogeneous Graph Masked Autoencoder, proposed by [Tian et al. \(2023\)](#) and presented as an oral paper at AAAI 2023, represents the state of the art in self-supervised pretraining on heterogeneous graphs. HGMAE extends the masked autoencoding paradigm to heterogeneous settings through three complementary training objectives: (i) metapath-based edge reconstruction, which reconstructs masked edges in metapath-derived adjacency matrices with semantic-level attention weighting; (ii) target attribute restoration, which recovers masked node features using learnable mask tokens with an adaptive masking schedule; and (iii) positional feature prediction, which captures structural roles through metapath2vec-based positional encodings. On standard heterogeneous graph benchmarks, HGMAE achieves 92.71 percent Micro-F1 on DBLP and 90.24 percent on ACM for node classification, competitive with or superior to fully supervised methods.

However, HGMAE has not been applied to financial transaction data or fraud detection, leaving open the question of whether its self-supervised heterogeneous representations transfer effectively to this domain.

2.8 Explainability in Graph-Based Fraud Detection

The question of explainability deserves particular emphasis given the regulatory environment in which fraud detection models operate. Banks must be able to justify their monitoring decisions to auditors and financial supervisory authorities, and a model that improves detection but cannot explain its reasoning may be unacceptable from a compliance perspective (Vallarino, 2025).

Gerlings and Constantiou (2023) documented a fundamental disconnect between the explanations that machine learning models provide and the explanations that investigators need. Their study found that investigators desired explanations grounded in the business domain—which counterparties were involved, what transaction patterns were unusual—rather than statistical performance summaries. Rao et al. (2022) addressed this partially through xFraud’s hybrid explainer, which combines GNNExplainer edge weights with graph centrality measures to identify the most informative edges in a prediction’s subgraph.

Heterogeneous GNNs offer a structural advantage for explainability: attention weights over different relation types can reveal which kinds of relationships—shared counterparties, unusual transaction chains, atypical geographic patterns—contributed most to a prediction, providing more topology-aware explanations than per-feature importance scores from tabular models.

2.9 Evaluation Under Extreme Class Imbalance

Real-world fraud detection datasets commonly exhibit positive rates well below one percent, meaning that a naive classifier labeling all transactions as legitimate can achieve over 99 percent accuracy. Accuracy is therefore a misleading metric, and the field has converged on metrics sensitive to performance on the minority class.

The area under the receiver operating characteristic curve (AUC-ROC) measures discriminative ability across all classification thresholds and is widely reported: xFraud achieved 0.91 on eBay data (Rao et al., 2022), while LaundroGraph reached 0.95 on real banking data (Cardoso et al., 2022). However, AUC-ROC can be overly optimistic under severe imbalance because it credits performance at high false positive rates that would be operationally unacceptable. The area under the precision-recall curve (AUC-PR) is arguably more informative, as it directly measures the tradeoff between precision and recall. Singh et al. (2024) reported AUC-PR of 0.89 on credit card fraud.

From an operational perspective, metrics beyond statistical performance also matter. Naser Eddin et al. (2021) emphasized false positive reduction as a key metric—their system reduced false positives by 80 percent—because every false positive represents wasted investigator time. Gerlings and Constantiou (2023) further noted that time-to-report is an operationally critical metric that is rarely reported in the academic literature but directly

affects regulatory compliance.

2.10 Summary and Research Gap

The literature reveals two parallel threads that have not yet been connected in the context of transaction-level fraud detection:

1. **Heterogeneous GNNs** (HGT, HMPNN) have demonstrated that preserving type information in financial graphs improves detection quality over homogeneous approaches, but these methods rely entirely on supervised learning and therefore inherit the limitations of scarce and incomplete fraud labels.
2. **Self-supervised graph learning** (LaundroGraph) has shown that useful representations can be learned without labels, but only on homogeneous bipartite graphs that discard the rich heterogeneous structure of real banking data.

No prior work applies heterogeneous self-supervised pretraining—specifically HGMAE—to financial transaction data. This thesis fills that gap by combining HGMAE pretraining with downstream heterogeneous classifiers (HGT, HMPNN) on real banking data, addressing both label scarcity and structural richness simultaneously. The experimental ladder (L0–L4) is designed to isolate the contribution of each component: tabular features, graph structure, heterogeneous typing, and self-supervised pretraining.

Chapter 3

Data

3.1 Data Collection

The dataset originates from Danske Bank’s internal retail payment transaction system. Two separate tables were provided under a data sharing agreement.

The first is a *transaction table* containing retail payment transactions recorded over . Each row represents a single payment event and carries full operational metadata: the sending and receiving accounts, the transaction amount and currency, the payment method and channel, timestamps, and a set of institutional routing codes.

FILL: exact date range and total transaction count

The second is a *fraud labels table* containing confirmed fraud cases identified through Danske Bank’s investigation process. This table uses a transaction identifier column whose name differs from the corresponding column in the transaction table, requiring a manual key match to establish the join.

FILL: description of sampling strategy and resulting dataset size.

The fraud labels table was then joined onto this subset on the matched transaction ID key, adding `CONFIRMED_RISK` as a single binary column: 1 for transactions present in the fraud labels table, 0 otherwise.

A note on label semantics. `CONFIRMED_RISK = 0` does not denote a *confirmed legitimate* transaction—it denotes *absence from the known fraud table*. Transactions with label 0 may include uninvestigated cases that are fraudulent but have not yet been identified. The true proportion of such cases is unknown. This distinction is consequential for model evaluation and is revisited in Section 4.8. It also motivates the choice of Area Under the Precision-Recall Curve (AUPRC) as the primary evaluation metric, rather than accuracy or AUROC, both of which assume that label 0 is a reliable negative.

3.2 Dataset Overview

Table 3.1 summarises the key properties of the working dataset.

Table 3.1: Dataset overview.

Property	Value
Total transactions	[FILL]
Time period covered	[FILL]
Transaction type	Retail payments
Confirmed fraud cases	[FILL (% rate)]
Unique sending accounts	[FILL]
Unique receiving accounts	[FILL]

The dataset is severely class-imbalanced. Under this imbalance, a model that predicts every transaction as legitimate achieves near-perfect accuracy while detecting no fraud whatsoever. Accuracy is therefore an uninformative metric throughout this work; AUPRC is used as the primary evaluation criterion.

3.3 Dataset Description

Rather than enumerate columns abstractly, this section walks through the dataset as a payment transaction, describing what each group of fields captures and how they relate to the fraud detection problem.

3.3.1 The Transaction Itself

Each row represents a single payment event initiated by a bank customer. The timestamp `EVENTTIME` records when the transaction was initiated; it is used to construct the temporal train/validation/test split described in Section 3.6.

The monetary value is recorded as `BASEVALUE` in a standardised reporting currency. Values are heavily right-skewed and are log-transformed ($\log(1 + x)$) prior to use as model features.

3.3.2 The Parties

The sending account is identified by `ACCOUNTID`, representing unique accounts belonging to Danske Bank’s own retail customers. The receiving account is identified by `COUNTERPARTYID`.

A striking structural property emerges from examining the overlap between these two pools: the vast majority of receivers are external counterparties that the bank has no direct record of as customers. This asymmetry reflects the nature of retail payment flows, where money predominantly moves outward from internal customers to external parties. It directly informs the graph construction described in Section 3.5, where sender and receiver accounts are modelled as structurally distinct node types.

3.3.3 Payment Classification

TRANSACTIONONUS is a binary flag indicating whether both the sending and receiving accounts are held at the same bank. External transactions carry a higher fraud rate than on-us transactions, motivating the treatment of on-us and external transactions as distinct edge types in the graph.

PAYMENTSUBMETHOD provides a finer-grained payment rail classification: *realTime*, *bankGiro*, *plusGiro*, *futurePayment*, *salary*, and minor residual categories. The Nordic giro rails are specific to Swedish and Finnish payment infrastructure; *futurePayment* covers scheduled future-dated transfers; salary transactions are payroll disbursements. PAYMENTCLEARING records the clearing speed; express and instant clearing reduces the window available for fraud interception.

INTERNATIONALFLAG is a binary indicator of cross-border transactions.

3.4 Feature Engineering

3.4.1 Edge Features (Transaction-Level)

Each transaction edge carries the following engineered features:

- $\log(1 + \text{BASEVALUE})$, z-score normalised
- One-hot encoding of CHANNEL
- One-hot encoding of PAYMENTSUBMETHOD
- Binary clearing express flag
- Binary international flag
- One-hot encoding of ACCOUNTBRANCH_TBE
- Cyclical time encoding: $\sin / \cos$ of hour-of-day and day-of-week (4 dimensions)

All z-score parameters and one-hot vocabularies are fitted exclusively on the training split.

3.4.2 Node Features (Aggregated from Training Data)

Node features are computed by aggregating transaction-level statistics from the training period only, preventing information leakage from future transactions.

InternalAccount features.

ExternalAccount features.

FILL: list c
rent feature
(out-degree,
amount stat
counterpart
diversity, etc

3.5 Graph Construction

The transaction data is represented as a heterogeneous directed graph, where accounts form nodes and transactions form edges. Three graph topology variants are explored, each encoding different levels of structural detail.

3.5.1 V1: Onus vs. External (2 Edge Types)

The simplest heterogeneous variant distinguishes only between on-us transfers (both accounts at the same bank) and external transfers:

- $\text{InternalAccount} \xrightarrow{\text{onus_transfer}} \text{InternalAccount}$
- $\text{InternalAccount} \xrightarrow{\text{external_transfer}} \text{ExternalAccount}$

3.5.2 V2: Payment-Rail-Typed Edges (6 Edge Types)

V2 preserves the on-us edge type but splits external transfers by payment rail:

- $\text{InternalAccount} \xrightarrow{\text{onus_transfer}} \text{InternalAccount}$
- $\text{InternalAccount} \xrightarrow{\text{ext_realtime}} \text{ExternalAccount}$
- $\text{InternalAccount} \xrightarrow{\text{ext_giro}} \text{ExternalAccount}$
- $\text{InternalAccount} \xrightarrow{\text{ext_future}} \text{ExternalAccount}$
- $\text{InternalAccount} \xrightarrow{\text{ext_salary}} \text{ExternalAccount}$
- $\text{InternalAccount} \xrightarrow{\text{ext_other}} \text{ExternalAccount}$

This variant tests whether preserving payment rail information in the graph topology improves detection over encoding it solely as an edge feature.

3.5.3 TXN_V1: Transactions as Nodes

The node-classification variant models transactions as first-class nodes rather than edges:

- $\text{InternalAccount} \xrightarrow{\text{sends}} \text{Transaction}$
- $\text{Transaction} \xrightarrow{\text{received_by_internal}} \text{InternalAccount}$
- $\text{Transaction} \xrightarrow{\text{received_by_external}} \text{ExternalAccount}$

This allows transaction-level features to participate in message passing as node attributes rather than edge attributes.

3.6 Temporal Split

Transactions are split chronologically to mirror real deployment conditions:

- **Training:** all transactions before
- **Validation:** transactions between
- **Test:** transactions after

L: first cut-
late

L: first and
nd cutoff

L: second

This ensures the model is always evaluated on transactions that postdate its training window, testing inductive generalisation to accounts and counterparties that may not have appeared during training.

3.7 Class Imbalance

The severe class imbalance is addressed through inverse-frequency weighting of the binary cross-entropy loss:

$$w_{\text{pos}} = \frac{n_{\text{negative}}}{n_{\text{positive}}}$$

This ensures that the gradient contribution of each fraud transaction is scaled proportionally to its rarity. The weighting is recomputed for each training split.

Chapter 4

Methodology

This chapter describes the experimental ladder — a systematic progression of models designed to isolate the contribution of graph structure, heterogeneous typing, and self-supervised pretraining to fraud detection performance.

4.1 Experimental Ladder

The experimental design follows a ladder of increasing structural complexity:

Level 0 — Tabular Baselines

Logistic Regression and XGBoost operating on transaction-level features only. Establishes the baseline performance achievable without any graph information.

Level 1 — Graph-Augmented Features

Hand-crafted graph features (degree statistics, neighborhood aggregates) fed to XGBoost. Tests whether graph structure carries signal beyond tabular features.

Level 2 — Homogeneous GNNs and KGE

GCN, GraphSAGE, TransE, and DistMult on a homogeneous transaction graph. Tests whether end-to-end learned representations outperform hand-crafted graph features.

Level 3 — Heterogeneous GNNs

HGT and HMPNN on heterogeneous graph variants (V1, V2, TXN_V1). Tests whether preserving type information improves detection quality.

Level 4 — Self-Supervised Pretraining

HGMAE and LaundroGraph for self-supervised pretraining, followed by fine-tuning with a downstream classifier. Tests whether self-supervised representations improve performance under label scarcity.

Each level is designed to answer a specific research question: L0→L1 asks whether graph structure helps at all; L1→L2 asks whether GNNs learn better representations than hand-crafted features; L2→L3 asks whether heterogeneous typing improves over homogeneous; and L3→L4 asks whether self-supervised pretraining adds value.

4.2 L0: Tabular Baselines

4.2.1 Logistic Regression

A regularised logistic regression model serves as the simplest baseline. The model operates on the same edge feature vector used across all levels (Section 3.4), applying an ℓ_2 -regularised linear classifier with inverse-frequency class weighting.

4.2.2 XGBoost

A gradient-boosted tree ensemble using the XGBoost implementation. The model uses the same feature vector as logistic regression, with `scale_pos_weight` set to the inverse class frequency ratio. Hyperparameters are tuned on the validation split.

4.3 L1: Graph-Augmented Features

Level 1 augments the tabular feature vector with hand-crafted graph features computed from the training-period transaction graph:

- Sender and receiver degree (in/out)
- Mean, standard deviation, and total transaction amount per account
- Counterparty diversity (number of unique counterparties)
- Channel diversity (entropy over payment channels)
- Temporal behaviour ratios (night/weekend transaction proportions)

These features are concatenated with the edge features and fed to an XGBoost classifier, allowing a direct comparison with L0 to isolate the contribution of graph-derived information.

4.4 L2: Homogeneous Models

4.4.1 Graph Convolutional Network (GCN)

The GCN baseline (Kipf and Welling, 2017) applies spectral graph convolutions on a homogeneous transaction graph. The layer-wise propagation rule is:

$$\mathbf{H}^{(l+1)} = \sigma\left(\tilde{\mathbf{D}}^{-\frac{1}{2}}\tilde{\mathbf{A}}\tilde{\mathbf{D}}^{-\frac{1}{2}}\mathbf{H}^{(l)}\mathbf{W}^{(l)}\right) \quad (4.1)$$

where $\tilde{\mathbf{A}} = \mathbf{A} + \mathbf{I}$ is the adjacency matrix with self-loops, $\tilde{\mathbf{D}}$ is the corresponding degree matrix, and σ is a nonlinearity.

For edge classification, source and target node embeddings are concatenated with the edge feature vector and passed through an MLP classifier.

4.4.2 GraphSAGE

GraphSAGE (Hamilton et al., 2017) replaces spectral convolution with neighbourhood sampling and aggregation:

$$\mathbf{h}_v^{(l+1)} = \sigma(\mathbf{W}^{(l)} \cdot \text{CONCAT}(\mathbf{h}_v^{(l)}, \text{AGG}(\{\mathbf{h}_u^{(l)} : u \in \mathcal{N}(v)\}))) \quad (4.2)$$

where AGG is a mean aggregator. GraphSAGE’s inductive design makes it suitable for the temporal split setting, where test nodes may not appear during training.

4.4.3 TransE

TransE (Bordes et al., 2013) learns entity and relation embeddings such that:

$$\mathbf{h} + \mathbf{r} \approx \mathbf{t} \quad (4.3)$$

for valid triples (h, r, t) , trained with a margin-based ranking loss. Edge classification is performed by scoring candidate triples and thresholding.

4.4.4 DistMult

DistMult (Yang et al., 2015) scores triples via a bilinear function:

$$f(h, r, t) = \mathbf{h}^\top \text{diag}(\mathbf{r}) \mathbf{t} \quad (4.4)$$

Like TransE, DistMult is trained with a contrastive objective and evaluated by scoring transaction edges.

4.5 L3: Heterogeneous GNNs

4.5.1 Heterogeneous Graph Transformer (HGT)

The HGT implementation follows [Hu et al. \(2020\)](#), using type-dependent linear projections for queries, keys, and values:

$$\mathbf{Q}_i^{(l)} = \mathbf{W}_{\tau(t)}^Q \mathbf{h}_t^{(l-1)} \quad (4.5)$$

$$\mathbf{K}_i^{(l)} = \mathbf{W}_{\tau(s)}^K \mathbf{h}_s^{(l-1)} \quad (4.6)$$

$$\mathbf{V}_i^{(l)} = \mathbf{W}_{\tau(s)}^V \mathbf{h}_s^{(l-1)} \quad (4.7)$$

where $\tau(\cdot)$ denotes the node type function. Attention weights are computed per meta-relation triplet $(\tau(s), \phi(e), \tau(t))$, allowing the model to learn type-specific importance patterns.

For edge classification, source and target node embeddings from the final HGT layer are concatenated with the edge feature vector and passed through an MLP edge classifier.

4.5.2 HMPNN

Following [Johannessen and Jullum \(2023\)](#), the Heterogeneous Message Passing Neural Network uses edge-type-specific message functions that incorporate edge features:

$$\mathbf{m}_{s \rightarrow t}^{(l)} = \text{MLP}_{\phi(e)}(\text{CONCAT}(\mathbf{h}_s^{(l)}, \mathbf{e}_{s,t})) \quad (4.8)$$

where $\phi(e)$ is the edge type and $\mathbf{e}_{s,t}$ is the edge feature vector. Messages are aggregated per edge type using mean pooling, then combined across types via summation before updating the target node embedding.

4.6 L4: Self-Supervised Pretraining

4.6.1 HGMAE

The Heterogeneous Graph Masked Autoencoder ([Tian et al., 2023](#)) is adapted for the transaction graph domain. The pretraining phase optimises three complementary objectives:

1. **Metapath-based edge reconstruction:** masked edges in metapath-derived adjacency matrices are reconstructed with semantic-level attention weighting.
2. **Target attribute restoration:** masked node features are recovered using learnable mask tokens with an adaptive masking schedule.

3. **Positional feature prediction:** structural roles are captured through metapath2vec-based positional encodings.

After pretraining, the encoder weights initialise a downstream HGT or HMPNN classifier that is fine-tuned on the labeled fraud data.

4.6.2 LaundroGraph

Following [Cardoso et al. \(2022\)](#), a self-supervised link prediction objective is applied to the transaction graph. A GAT encoder learns node embeddings by predicting held-out edges, and the resulting representations are used as input features for a downstream classifier.

4.6.3 Motivation for Self-Supervised Pretraining

The fraud detection setting exhibits acute label scarcity: confirmed fraud accounts for approximately % of all transactions, meaning that a supervised binary cross-entropy objective is overwhelmingly dominated by the legitimate class. The encoder receives meaningful gradient signal from fraud examples on only a small fraction of training steps, limiting its ability to learn discriminative structural patterns. Self-supervised pretraining addresses this imbalance by allowing the encoder to learn from *all* transactions equally, without label bias, before any fraud-specific fine-tuning.

FILL

Three lines of prior work motivate the L4 design. First, [Rao et al. \(2022\)](#) demonstrate that a Heterogeneous Graph Transformer applied to a multi-typed transaction graph achieves strong fraud detection performance at production scale, validating HGT as an appropriate encoder architecture for this domain. Second, autoencoder-based approaches to fraud detection ([Pumsirirat and Yan, 2018](#); [Singh et al., 2024](#)) show that self-supervised reconstruction objectives can capture anomalous patterns in financial data — fraudulent transactions produce higher reconstruction error because the encoder has learned the distribution of legitimate behaviour. Third, [Tian et al. \(2023\)](#) introduce Heterogeneous Graph Masked Autoencoders, combining masked autoencoding with heterogeneous graph structure and demonstrating state-of-the-art performance on node classification benchmarks through type-aware masking and reconstruction.

The contribution of L4 lies in combining all three ideas: HGMAE-style pretraining on a heterogeneous transaction graph, followed by supervised fine-tuning for fraud detection. Crucially, the experimental ladder provides a controlled comparison — if L4 outperforms L3 using the *same* encoder architecture and graph structure, the performance gain is directly attributable to the self-supervised pretraining stage rather than to architectural or data-representation differences.

FILL: finalise
L4 impleme
tion details
they stabilis

4.7 Training Protocol

All neural models share a common training protocol:

- **Optimiser:** Adam with default parameters ($\beta_1 = 0.9$, $\beta_2 = 0.999$)
- **Learning rate:**
- **Loss:** Binary cross-entropy with inverse-frequency class weighting
- **Early stopping:** patience of epochs on validation PR-AUC
- **Batch:** full-batch training (graph fits in memory)

L: initial
and sched-

L

4.8 Evaluation Framework

The primary evaluation metric is the **Area Under the Precision-Recall Curve (PR-AUC)**, which directly measures the tradeoff between precision and recall under extreme class imbalance. Secondary metrics include:

- **AUC-ROC:** discriminative ability across all thresholds
- **F1 score:** harmonic mean of precision and recall at optimal threshold
- **False positives per true positive:** operational cost metric

All metrics are computed on the temporally held-out test split. Statistical significance is assessed where applicable.

Chapter 5

Experiments and Results

5.1 Experimental Setup

Hardware, software version, random seed, reproducibility details.

5.2 L0 and L1 Results: Tabular and Graph Feature Baselines

Report L0 and L1 results.

5.3 L2 Results: Homogeneous GNNs and KGE

Report GCD, GraphSAGE, TransE, Disjoint, Mult results.

5.4 L3 Results: Heterogeneous GNNs

5.5 L4 Results: Self-Supervised Pretraining

Report HG and HMPN results across V1, V2, TXN_V1 topologies.

5.6 Topology Ablation: V1 vs. V2

5.7 Cross-Level Comparison

Report HG-MAE and LaundroGra pretraining fine-tuning results.

5.8 Explainability Analysis

Compare graph topology variants.

Unified comparison table across all levels addressing

Chapter 6

Discussion

6.1 Hypothesis Assessment

Evaluate H1, H2, H3, and H4 against experimental evidence.

6.2 Practical Implications for Fraud Detection Operations

Discuss operational impact of false positive reduction, investigator workload.

6.3 Limitations

6.4 Comparison with Prior Work

Label noise, single bank, no temporal drift evaluation, computational cost.

Position results relative to HMPNN on DNB data, LaundroGra, xFraud.

Chapter 7

Conclusion

7.1 Summary of Findings

Summarise findings from the experimental ladder.

7.2 Future Work

Temporal dynamics, continuous learning, cross-batch generalisation, deployment considerations

Appendix A

Feature Reference

Complete
feature ta-
bles: edge fe-
tures, node
features per
type, dropped
columns with
rationale.

Appendix B

Hyperparameter Settings

Complete hyperparameter settings tables for all models (L0-L4).

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